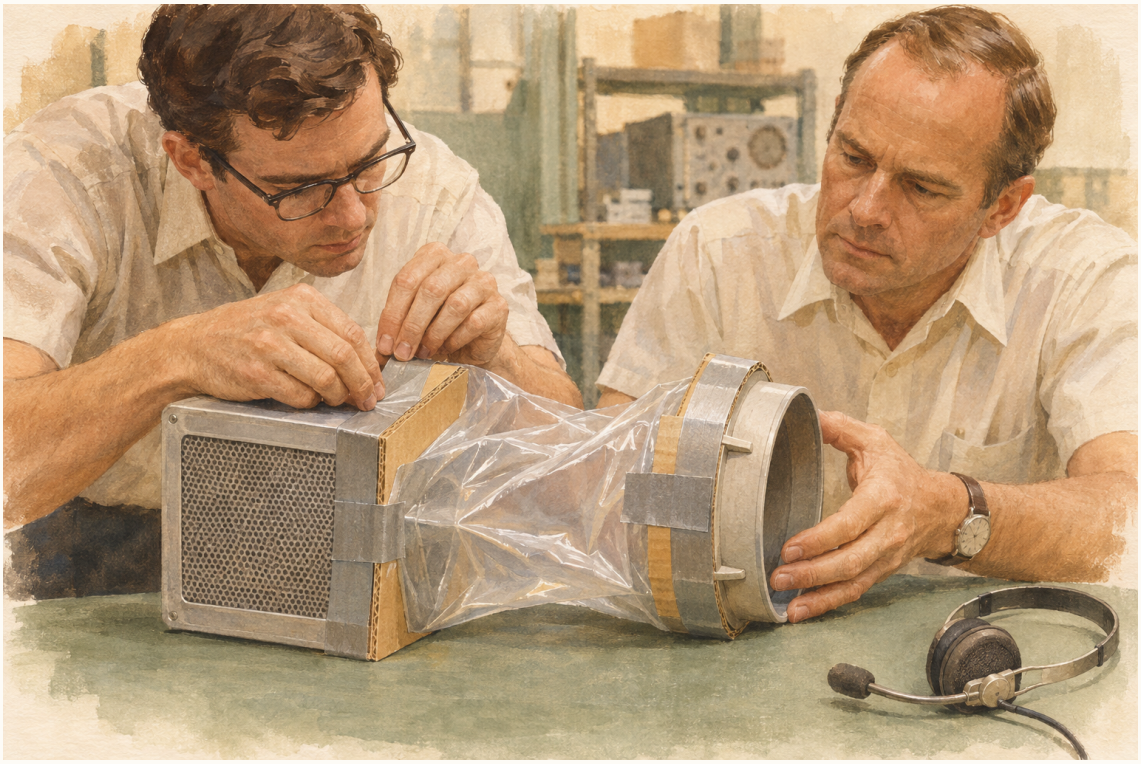
阿波罗13号：成功的失败

Apollo 13: The Successful Failure



方形滤芯怎样接进圆形插槽，救回三名宇航员？

英文原文

On April 11th 1970 NASA **launched** Apollo 13, meant for the third **crewed lunar landing**. On April 13th, some 200,000 miles out, an **oxygen tank exploded** in the **service module**, home to most of the ship's oxygen and power. The **blast crippled** the **spacecraft**; the landing was **aborted**.

The **astronauts took refuge in** the **lunar module**, Aquarius, built to support two men for about two days. Using its **engine**, the crew **steered** the **craft onto** a **trajectory** around the Moon, whose gravity **swung** it home. They **rationed** water, **conserved power**, and **shivered** in a **bitterly** cold **cabin**.

A quieter **hazard emerged**: with a third man aboard, **carbon dioxide** was **accumulating**. The command module, the crew's **capsule**, used square **filter cartridges**; Aquarius's **sockets** were round. From plastic bags, **cardboard**, and **tape** on board, ground engineers **devised** an **adapter**, then **talked** the crew **through assembling** it. On April 17th the capsule **splashed down** in the Pacific.

Yet the **fault** had been **built in** on Earth. A NASA **review board traced** the **explosion to** the tank's **thermostat switches**, designed for the ship's 28-**volt** supply. In 1965 the spacecraft's **builder rewrote** the **specification** to let the **heaters run on** 65-volt launch-site power; the switches were never **upgraded**, a **mismatch** neither that **contractor** nor NASA **detected**. Weeks before launch, to **drain** the **stubborn** tank, a **heater** ran about eight hours. The switches probably **malfunctioned**; **unchecked** heat **scorched** the **insulation** on the tank's **wiring**; in flight, that insulation **ignited**.

NASA called Apollo 13 a "successful failure". The **paradox** holds: a flight **undone by** a five-year-old engineering **oversight** was **salvaged** within four days by **ingenuity** on the ground.

中文译文

1970年4月11日，NASA发射了阿波罗13号，原本是要完成第三次载人的登月。4月13日，大约离地球200,000英里远的地方，一个氧气罐爆炸了，发生在服务舱段里面，存储着大部分飞船的氧气和电力。这股爆炸气浪严重毁坏了宇宙飞船；登月被终止了。

宇航员们躲进登月舱水瓶座号避难，它被建造出来原本是为了支持两个人生存大约两天左右。使用它的发动机/引擎，机组操纵航天器驶入一条绕月轨迹，月球的引力把它甩回了地球。他们限定用水量，节约用电，并且不停地打哆嗦，身处极其寒冷的舱室里。

一个悄无声息的隐患出现了：随着第三个人登舱，二氧化碳正在积聚。指令舱，即机组太空舱，用的是方形的过滤器滤芯；然而水瓶座号的插槽却是圆的。用飞船上现有的塑料袋、硬纸板和胶带，地面工程师构思设计出一个转接器/转换插头，然后一步步指导机组把它组装起来。4月17日，太空舱水上降落在太平洋。

然而，这个缺陷早在地球上就被内置了/埋下了。一个NASA的审查委员会查明了这次爆炸源自于（液氧）罐的恒温器开关上，这些开关是按飞船的28伏电源设计的。在1965年，飞船承建商就改写了技术规格，允许加热器使用发射场65伏的电源来运行；开关却始终没有升级，这处不匹配，无论是那家承包商还是NASA都没有发现。发射前几周，为了排空那个难以排空的储罐，一台加热器连续工作了约八个小时。开关很可能失灵了；无人管控的高温烧焦了罐内线路上的绝缘层；在飞行途中，这层绝缘材料燃烧起来了/着火了。

NASA把阿波罗13号称作“成功的失败”。这个悖论站得住脚：一次飞行，毁于五年前的一处工程疏忽，却在四天之内得以挽救，靠的是地面上的巧思/足智多谋。

词汇与表达

launched v.
发射了

crewed adj.
载人的

lunar landing adj.
登月

oxygen tank n.
氧气罐

exploded v.
爆炸了

service module n.
服务舱段

blast n.
爆炸气浪

crippled v.
严重毁坏了
The blast crippled the spacecraft

spacecraft n.
宇宙飞船

aborted v.
终止了
the landing was aborted

astronauts n.
宇航员

took refuge in n.
躲进……避难
The astronauts took refuge in the lunar module, Aquarius,

lunar module adj.
登月舱

engine n.
发动机/引擎

steered onto v.
操纵……驶入

craft n.
航天器

trajectory n.
轨迹
a trajectory around the Moon

swung v.
甩回了

rationed v.
限定……量
They rationed water

conserved power v.
节约用电

shivered v.
打哆嗦

bitterly adv.
极其

cabin n.
舱室

hazard n.
隐患
A quieter hazard emerged

词汇与表达（续）

emerged v.
出现了
A quieter hazard emerged

carbon dioxide n.
二氧化碳

accumulating v.
积聚
carbon dioxide was accumulating

capsule n.
太空舱

filter n.
过滤器

cartridges n.
滤芯

sockets n.
插槽

cardboard n.
硬纸板

tape n.
胶带

devised v.
构思设计出
ground engineers devised an adapter

adapter n.
转接器/转换插头
devised an adapter

talked through v.
一步步指导

assembling v.
组装
through assembling it

splashed down v.
水上降落

fault n.
缺陷
the fault had been built in on Earth

built in v.
内置了/埋下了

review board n.
审查委员会

traced to v.
查明了……源自于

explosion n.
爆炸

thermostat n.
恒温器
the tank's thermostat switches

switches n.
开关

volt n.
伏

词汇与表达（续）

builder n.
承建商

rewrote v.
改写了

specification n.
技术规格
rewrote the specification

heaters n.
加热器

run on v.
使用……的电源来运行

upgraded v.
升级

mismatch n.
不匹配
a mismatch

contractor n.
承包商

detected v.
发现

drain v.
排空
to drain the stubborn tank

stubborn adj.
难以排空的

heater n.
加热器

malfunctioned v.
失灵了
The switches probably malfunctioned

unchecked adj.
无人管控的

scorched v.
烧焦了

insulation n.
绝缘层
the insulation on the tank's wiring

wiring n.
线路

ignited v.
燃烧起来了/着火了

paradox n.
悖论
The paradox holds

undone by v.
毁于

oversight n.
疏忽
a five-year-old engineering oversight

salvaged v.
挽救
was salvaged within four days

ingenuity n.
巧思/足智多谋
by ingenuity on the ground

文章简述

- 事故：1970年，阿波罗13号服务舱的氧气罐爆炸，登月取消。三名宇航员躲进原本只供两人使用约两天的登月舱。
- 险情：他们缺水缺电、忍受严寒，舱内二氧化碳还不断积聚；能清除它的方形滤芯，却接不上登月舱的圆形接口。
- 救急：地面工程师用船上已有的塑料袋、纸板和胶带设计转接器，再指导机组装好，解决滤芯不匹配的问题。
- 根源：氧气罐加热器的规格已改为兼容发射场的65伏电源，罐内温控开关却仍按飞船的28伏电源设计。发射前长时间加热时，开关可能失灵，烧伤电线绝缘层，后来在飞行中起火。
- 返航：机组用登月舱发动机调整路线，借月球引力绕回地球，爆炸四天后安全溅落。没登成月，却救回三人，故称“成功的失败”。